Ruijia (Regina) Cheng

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Research Interests: Human-computer Interaction; Responsible AI & Safety; Developer tools; Creativity support; End-user programming; Data literacies; Social computing; Learning technology

Education

2023 **University of Washington**
Doctor of Philosophy in Human Centered Design & Engineering
Committee: Benjamin Mako Hill, Jennifer Turns, Sayamindu Dasgupta, Amy Zhang
Dissertation: *Data Literacies in Informal Settings*

2021 **University of Washington**
Master of Science in Human Centered Design & Engineering

2018 **University of California, San Diego**
Magna Cum Laude
Bachelor of Science in Cognitive Science with a Specialization in Computation
Bachelor of Science in Mathematics: Applied Science

Experiences

07/23 – present **Apple | Human Centered Machine Intelligence (HCMI)** Seattle, WA
Senior Research Scientist (10/26 – present)
Research Scientist (09/24 – 09/26)
AIML Resident (07/23 – 08/24)

Leads research and product development on human-computer interaction, responsible AI, UI understanding, accessibility, and creativity support.

09/18 – 06/23 **University of Washington | Human Centered Design & Engineering** Seattle, WA
Graduate Research & Teaching Assistant.

Led mixed-method research projects to understand and design for end-user programmers across several settings: online communities, data science collaboration and communication, visual block-based programming systems for data literacy, and creative feedback exchange.

06/22 – 09/22 **Microsoft Research | Software Analysis and Intelligence in Engineering Systems**
Redmond, WA
PhD Research Intern.

Led a multi-phase project combining interviews, prototyping, and a design probe to explore sociotechnical affordances for developers to build trust in code generation tools. Collaborated on research on responsible AI interface design and trust in software development.

09/21 – 12/21 **Dataminr | Research and Innovation** New York City, NY
PhD Research Intern.

Led and collaborated on research into human-in-the-loop text summarization. Conducted a large-scale systematic literature review, developed interactive prototypes, ran design probe interview studies with crowd workers, and established a taxonomy for human-AI collaboration in text summarization.

06/21 – 09/21 **Northwestern University | Community Data Science Collective** Evanston, IL
Visiting Researcher.

Led a large-scale quantitative study of data literacy and social media discussion about COVID-19, building datasets of cross-platform social media activity to support the analysis.

03/21 – 06/21 **Microsoft | Education via i2e LLC** Redmond, WA
Project Intern.

Designed and developed K-12 curricula, user scenarios, and interaction guides for Search Coach, a K-12 education product.

06/20 – 09/20 **Facebook | Watch** Menlo Park, CA
UX Research Intern.

Designed and conducted usability tests, in-app surveys of 20k+ responses across 5 countries, and analyses of 20k+ user logs, working with cross-functional teams (engineering, design, and data) and vendors. The findings shaped the design of video platforms and players.

10/16 – 01/18 **University of California, San Diego | Design Lab** La Jolla, CA
Undergraduate Research Assistant.

Led survey and online experiment studies on crowd creativity and problem framing, and conducted thematic analyses and topic modeling of narrative patterns in computational notebooks.

Publications

Journal Publications

[J2] R. Cheng, A. Shaw, and B. M. Hill. "Conspiracist attributes differentiate pro- and anti-vaccine online discourses about data". In: *Scientific Reports* (2026). Accepted, to appear.

[J1] R. Cheng, R. Wang, T. Zimmermann, and D. Ford. ""It would work for me too": How online communities shape software developers' trust in AI-powered code generation tools". In: *ACM Transactions on Interactive Intelligent Systems* 14.2 (2024), pp. 1–39.

Peer-reviewed Conference Proceedings

[C16] J.-T. Beason, R. Cheng, E. Schoop, and J. Nichols. "Athena: Intermediate Representations for Iterative Scaffolded App Generation with an LLM". In: *Proceedings of the 31st International Conference on Intelligent User Interfaces*. 2026, pp. 284–306.

[C15] R. Cheng, J. T. Liang, E. Schoop, and J. Nichols. "Mapping the design space of user experience for computer use agents". In: *Proceedings of the 31st International Conference on Intelligent User Interfaces*. 2026, pp. 646–662.

[C14] H. Agrawal, E. Schoop, X. Pan, A. Mahajan, A. Seff, D. Feng, R. Cheng, A. R. M. Y. Teran, E. Gomez, A. Sundararajan, et al. "UINavBench: A framework for comprehensive evaluation of interactive digital agents". In: *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE. 2025, pp. 23353–23363.

[C13] A. Leung, R. Cheng, J. Wu, J. Nichols, and T. Barik. "SQUIRE: Interactive UI authoring via slot query intermediate representations". In: *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. 2025, pp. 1–17.

[C12] T. Tseng, R. Cheng, A. M. McNutt, and J. Nichols. "Keyframer: A Design Probe for Exploring LLM Assistance in 2D Animation Design". In: *2025 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE. 2025, pp. 33–45.

[C11] R. Cheng, T. Barik, A. Leung, F. Hohman, and J. Nichols. "BISCUIT: Scaffolding LLM-generated code with ephemeral UIs in computational notebooks". In: *2024 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE. 2024, pp. 13–23.

[C10] M. Taeb, A. Swearngin, E. Schoop, R. Cheng, Y. Jiang, and J. Nichols. "Axnav: Replaying accessibility tests from natural language". In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 2024.

[C9] R. Wang, R. Cheng, D. Ford, and T. Zimmermann. "Investigating and designing for trust in ai-powered code generation tools". In: *Proceedings of the 2024 ACM conference on fairness, accountability, and transparency*. 2024, pp. 1475–1493.

[C8] R. Cheng, A. Dangol, F. M. T. Ello, L. Wang, and S. Dasgupta. "Concepts, practices, and perspectives for developing computational data literacy: Insights from workshops with a new data programming system". In: *Proceedings of the 22nd annual ACM interaction design and children conference*. 2023, pp. 100–111.

[C7] R. Cheng, S. Dasgupta, and B. M. Hill. "How interest-driven content creation shapes opportunities for informal learning in Scratch: A case study on novices' use of data structures". In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 2022, pp. 1–16. **Best Paper Honorable Mention Award (Top 5%)**.

[C6] R. Cheng and J. Frens. "Feedback exchange and online affinity: A case study of online fanfiction writers". In: *Proceedings of the ACM on Human-Computer Interaction* 6.CSCW2 (2022), pp. 1–29. Authors contributed equally.

[C5] R. Cheng and B. M. Hill. "Many destinations, many pathways: A quantitative analysis of legitimate peripheral participation in scratch". In: *Proceedings of the ACM on human-computer interaction* 6.CSCW2 (2022), pp. 1–26.

[C4] R. Cheng, A. Smith-Renner, K. Zhang, J. R. Tetreault, and A. Jaimes. "Mapping the design space of human-ai interaction in text summarization". In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics. 2022, pp. 431–455.

[C3] V. Lai, A. Smith-Renner, K. Zhang, R. Cheng, W. Zhang, J. Tetreault, and A. Jaimes. "An exploration of post-editing effectiveness in text summarization". In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics. 2022, pp. 475–493.

[C2] R. Cheng and M. Zachry. "Building community knowledge in online competitions: Motivation, practices and challenges". In: *Proceedings of the ACM on Human-Computer Interaction* 4.CSCW2 (2020), pp. 1–22.

[C1] R. Cheng, Z. Zeng, M. Liu, and S. Dow. "Critique me: exploring how creators publicly request feedback in an online critique community". In: *Proceedings of the ACM on Human-Computer Interaction* 4.CSCW2 (2020), pp. 1–24.

Preprints

[P3] Y. Ning, R. Cheng, S. Ghosh, P. Duan, H. Lin, A. Sundararajan, S. Pandey, H. Sun, J. Nichols, and E. Schoop. "iPrefBench: A User-Centric Benchmark for Personalized Mobile Use Agents". In: *arXiv preprint arXiv:2610.34735* (2026).

[P2] J. T. Liang, T. Barik, J. Nichols, E. Schoop, and R. Cheng. "Agentbuilder: Exploring scaffolds for prototyping user experiences of interface agents". In: *arXiv preprint arXiv:2510.04452* (2025).

[P1] T. Tseng, R. Cheng, and J. Nichols. "Keyframer: Empowering animation design using large language models". In: *arXiv preprint arXiv:2402.06071* (2024).

Short Papers, Posters, and Workshop Papers

[S3] R. Cheng, A. Smith-Renner, K. Zhang, J. R. Tetreault, and A. Jaimes. "Trust and Reliance in Human-AI Collaborative Text Summarization". In: *Workshop on Trust and Reliance in Human-AI Teams, CHI 2022*. 2022.

[S2] R. Cheng, J. De Castro, S. Dow, and J. Chan. "An Exploratory Study of Problem Framing in Distributed Collaborative Design". In: *Companion of the 2018 ACM Conference on Supporting Groupwork (GROUP)*. 2018.

[S1] F. Singh, A. Smith, N. Dudeck, E. Herrera, J. Lee, Z. Yang, R. Cheng, and J. Pineda. *A Pilot Study to Assess the Effects of EEG-Gamma Neurofeedback on Working Memory in Schizophrenia Patients*. Poster, Society for Neuroscience Annual Conference (SfN). 2016.

Awards & Honors

2024	HCDE PhD Research Award
2023	EECS Rising Stars
2023	Special Recognition of Outstanding Reviews, CHI 2024
2022	Best Paper Honorable Mention Award, CHI 2022
2020	Special Recognition of Outstanding Reviews, CSCW 2020
2014–18	Provost Honor, University of California, San Diego

Selected Talks

2025 "Supporting developers and designers in the era of generative AI." Invited speaker at the Informatics Speaker Series, University of California, Irvine.

2023 "Supporting the public to learn about and work with data and AI." Invited speaker at the EECS Rising Stars seminar. Atlanta, GA.

2023 "Understanding and designing for community-supported programming." Presentation in the department of Information Science & Technology, George Mason University. Fairfax, VA.

2022 "Supporting Computational Learning in Online Communities." Presentation at Microsoft MakeCode. Microsoft Research. Webinar.

2022 "Understanding and Supporting Informal Learning in Online Communities." Invited speaker at The Expertise@scale Salon. Emory University. Webinar.

2022 "All Communities Are Learning Communities." Invited speaker at The Science of Community Dialogues. Community Data Science Collective. Webinar.

2021 "Data Scientists or Conspiracists: Critical Discourses about COVID Data among Pro- and Anti-vaccine Tweets." Presentation at the HCDE research seminar (Autumn 2021). University of Washington. Webinar.

2021 "Imagining Future Design of Tools for Youth Data Literacies." Workshop organizer at the Connected Learning Summit 2021. Webinar.

2019 "Feedback-Seeking in Online Fanfiction Communities." Poster presentation at the 2019 HCDE Research Showcase. University of Washington. Seattle, WA.

2017 "Plug-N-Talk: An Affordable Solution to Hearing Loss." Finalist presentation at the 2nd UCSD ECE Annual Design Competition. University of California, San Diego. San Diego, CA.

Teaching

Guest Lectures

2024 "Understanding and Supporting Informal Learning in Online Communities." COM 481 and COM 581, University of Washington.

2019, 20 "A Crash Course on Statistics for Usability Testing." HCDE Usability Testing, University of Washington.

Directed Research Group

2022 "Evaluative Study on Dataland: Supporting Novices to Analyze Data." University of Washington.

2021, 22 "Supporting Critical Capacities in Data Science through Online Interactions." University of Washington.

Teaching Assistant

2020, 21 HCDE Capstone. University of Washington. Students won Best Design & Engineering awards.

2020, 21 HCDE Capstone Project Planning. University of Washington.

2020 HCDE Qualitative Methods. University of Washington.

2019 HCDE Usability Testing. University of Washington.

2019 HCI+D Formative UX Research Studio. University of Washington.

Mentoring

2026 Kyzyl Monteiro, Carnegie Mellon University. PhD research intern at Apple.

2025 Jenny T. Liang, Carnegie Mellon University. PhD research intern at Apple.

2023 Cindy Gong, University of Washington. Undergraduate Honor Thesis.

2019 Ziwen Zeng, University of California, San Diego. Undergraduate summer intern.

2019	Maysnow Liu, University of California, Davis. Undergraduate summer intern.

Academic Services

2026	Program Committee, ACM CSCW
2024–26	Associate Chair, ACM CHI
2024	Associate Chair, ACM DIS
2023	Program Committee, ACM FAccT
2022	Student Ambassador, UW DUB
2021	Doctoral Colloquium Organizer, UW DUB
2021	Workshop Organizer, Connected Learning Summit
2020	Application Reviewer, UW HCDE Master program
2019	Workshop Organizer, Community Data Science Collective
2019–present	Reviewer for ACM CHI, ACM CSCW, ACM UIST, ACM IDC, ACM EICS, ACM Creativity & Cognition, ACM TiiS, ACM TOCE, Scientific Reports

Professional References

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Denae Ford Robinson, denae@microsoft.com
Thomas Zimmermann, tzimmer@uci.edu